

Lab 10: Quantum Optics on the Bench

Single photons on a tabletop kit

1 ECTS · in person at i-LAMP

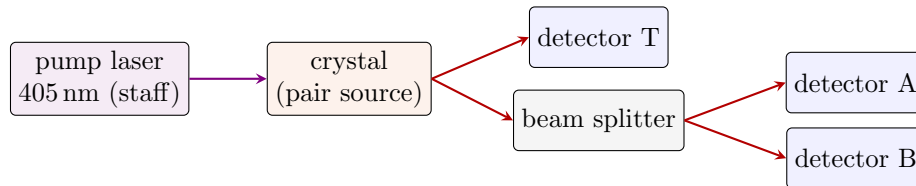
Prerequisites: *Fundamental Concepts of Quantum Technologies* (lecture series) — nothing else.

What you need: the Thorlabs EDU-QOP1 Quantum Optics Kit with its time tagger and software (set up by the staff); the worksheet (pages 4–5); a pen and a calculator.

Time: about 10 h, indicative: Part A 2 h, B 1.5 h, C 2 h, D 2.5 h, optional parts 2 h.

Source: Thorlabs, *EDU-QOP1(/M) Quantum Optics Kit User Guide*, Rev. D (2025-10-25), “manual §x”; follow its sections, this guide does not repeat them.

Goal: decide, from detector clicks alone, whether the light on the bench comes as single photons, and whether one photon can interfere with itself.



Parts B–D. A: dimmed 635 nm alignment laser into the beam splitter, no T. D: Michelson before B.

Safety first

The room is dark during measurements. The 405 nm pump laser is class 3B: **only the staff switch it on** (key, interlock). While it is on (Parts B–D) you follow the i-LAMP eyewear and conduct rules of the laser safety officer; the manual is explicit: “Always wear laser safety glasses when working with the pump laser!” (manual §7.6.4). You use the 635 nm alignment laser and the software (manual Ch. 1, Ch. 14).

Words you need

Beam splitter: a half-transparent mirror with two outputs (course document QC10, *Light*). **Click:** one pulse of a single-photon detector. **Coincidence:** two (or three) detectors click in the same **window** Δt (5 ns here, set in the software). **$g^{(2)}$:** with $P(X)$ the probability that detector X clicks in one window, $P(X) = R_X \Delta t$ for a click rate R_X (clicks per second),

$$g^{(2)} = \frac{P(A \text{ and } B)}{P(A)P(B)} : \quad 1 = \text{no correlation}, \quad < 1 = \text{fewer coincidences than chance}, \quad > 1 = \text{more}.$$

Accidentals: the coincidences expected by pure chance if A and B click independently, rate $R_A R_B \Delta t$; so $g^{(2)} = \text{coincidences} / \text{accidentals}$. **Poisson light:** photons arriving independently at random times, like raindrops; a laser, however dim, is Poisson light. **Pair source:** a crystal that, pumped at 405 nm, emits two photons (about 810 nm) at once. **Heralding:** a click at T announces that its partner photon is on the way to the beam splitter; A and B are counted only together with T. **Interference:** a wave split in two and recombined adds up or cancels, depending on the path difference; **visibility** $\mathcal{V} = (R_{\max} - R_{\min}) / (R_{\max} + R_{\min})$, 1 for perfect fringes. **Standard deviation** σ : the typical scatter of repeated results; a count of N clicks scatters by about $\sqrt{N}$.

Part A — Is a very dim laser a single-photon source?

A single photon cannot split: at the beam splitter it reaches A *or* B, never both, so a true single-photon source gives $g^{(2)} = 0$. Poisson light is different: each photon picks its exit on its own, so A and B click independently and $g^{(2)} = 1$ exactly, at *every* brightness. The alignment laser is dimmed by the kit’s filter; you look at A and B (Hanbury Brown–Twiss, HBT).

Formula → Number → Bench

$g_{\text{HBT}}^{(2)} = \frac{R_{AB}}{R_A R_B \Delta t}$. Manual §9.1, Table 1 ($\Delta t = 5$ ns): accidentals $R_A R_B \Delta t = 617.9$ Hz, measured $R_{AB} = 619.1$ Hz, so $g_{\text{HBT}}^{(2)} = 1.0020$ (the manual prints 1.00201).

Predict → measure → compare (worksheet W1)

Predict $g^{(2)}$. **Measure** following manual §9.1 (HBT tab, §11.7): its 20 s measurement, three times at the kit's setting. **Compare** with 1, with Table 1 (1.0020), and run to run. Dimmer light only enlarges the error bar (manual fn. 84): try it in the pre-lab playbox.

Counterintuitive — and Real

Manual §9.1 estimates how many photons are inside the 1 m setup at any moment: $n = lf/c = 1 \text{ m} \times 1.2 \cdot 10^6 \text{ s}^{-1} / 3 \cdot 10^8 \text{ m s}^{-1} = \mathbf{0.004}$ photons. Far below one — surely single photons? Yet $g^{(2)} \approx 1$ (Table 1: 1.0020), not 0. Rare photons are not single photons: a dim laser stays Poisson.

Part B — Does the crystal really make pairs?

The staff switch on the pump. If the crystal makes pairs, T and A click together far more often than by chance.

Formula → Number → Bench

$g_{\text{PS}}^{(2)} = \frac{R_{TA}}{R_T R_A \Delta t}$ = coincidences / accidentals. Manual §9.4, Table 2 (rates = counts / 20 s): $R_T = 231900 \text{ Hz}$, $R_A = 145250 \text{ Hz}$, $R_{TA} = 19293.25 \text{ Hz}$, so $g_{\text{PS}}^{(2)} \approx 115$: coincidences are $\approx 115 \times$ the accidentals. Table 2 prints no Δt ; we assume the standard $\Delta t = 5 \text{ ns}$ (manual §7.4.5).

Predict → measure → compare (worksheet W2)

Predict: the manual expects $g_{\text{PS}}^{(2)} \geq 30$ (§9.2). **Measure** following manual §9.2: Alignment tab (manual §9.2 calls it Adjustment; §11.5) or GRA tab (§11.7), detectors T and A. **Compare** with 30 and with Table 2.

Part C — Can one photon take both exits?

Now the partner photon, heralded by T, meets the beam splitter (Grangier–Roger–Aspect, GRA). Any classical light obeys $g^{(2)} \geq 1$ (manual eq. 75); one photon gives 0. The software counts a triple as a *coincidence of coincidences*: T&A and T&B both coincident (manual §4.2.2, §12.2.1).

Formula → Number → Bench

$g_{\text{GRA}}^{(2)} = \frac{N_{TAB} N_T}{N_{TA} N_{TB}}$ (counts: the measurement time cancels). Manual §9.4, Table 2 (20 s): $N_T = 4.638 \text{ M}$, $N_{TA} = 385865$, $N_{TB} = 324688$, $N_{TAB} = 513$, so $g_{\text{GRA}}^{(2)} = 0.0190$. With $\sigma \approx g_{\text{GRA}}^{(2)} / \sqrt{N_{TAB}} = 0.00084$, it lies $(1 - g_{\text{GRA}}^{(2)}) / \sigma \approx 1170$ standard deviations below the classical bound.

Predict → measure → compare (worksheet W3)

Predict: $g_{\text{GRA}}^{(2)} < 1$. **Measure** following manual §9.4 (GRA tab, §11.7): four counts and the time. **Compare:** how many σ below 1? Is the residual chance (next box)?

The bench is not the model: the window Δt

Detectors and time tagger resolve time only to about a nanosecond, so coincidences are counted in a window, and unrelated clicks in the same window count too. Two chance types fake triples (manual eqs. 105–106): $R^{(3)} = R_T R_A R_B \Delta t^2$ and $R^{(2+1)} = (R_{TA} R_B + R_{TB} R_A) \Delta t$, giving $g_{\text{acc}}^{(2)} = (R^{(3)} + R^{(2+1)}) R_T / (R_{TA} R_{TB})$. Table 2: $g_{\text{acc}}^{(2)} = 0.0193$, that is ≈ 520 chance triples in 20 s against 513 measured, a difference of 0.33σ . **The residual $g_{\text{GRA}}^{(2)}$ is consistent with chance coincidences alone: no sign of the photon splitting.** A shorter window or weaker pump lowers it (eq. 110).

The bench is not the model: real detectors

Our model is idealized. Detector efficiency multiplies the top and the bottom of every $g^{(2)}$ alike and cancels; dark counts and stray light enter only as accidentals.

Part D — Does a single photon interfere with itself?

The staff place a Michelson interferometer in one beam-splitter output, with B behind it: a second beam splitter sends the photon into two arms, mirrors send it back, and the arms recombine. The stage moves one mirror by x ; the light goes out and back, so the path difference is $2x$. The detector-side probability is $P_d(x) = (1 - \cos(4\pi x/\lambda))/2$, so R_{TB} rises and falls with period $\lambda/2$ in mirror position, while $g_{\text{GRA}}^{(2)} < 1$ along the scan.

Formula → Number → Bench

With $\lambda \approx 810$ nm the fringe period is ≈ 405 nm in mirror position (manual §11.9: “about 400 nm”). Backwards, one maximum per fringe: $\lambda = 2 \times (\text{mean spacing of maxima}) = 2(x_{\text{last}} - x_{\text{first}})/(N_{\text{max}} - 1) \approx 810$ nm, with a stage error of up to 3 % (manual §9.7).

Predict → measure → compare (worksheet W4)

Predict the period and λ . **Measure** following manual §9.7 (Michelson tab, §11.9); read $g_{\text{GRA}}^{(2)}$ during the scan. **Compare** λ with $810 \text{ nm} \pm 3\%$, and $\mathcal{V}$ with 1. Is $g_{\text{GRA}}^{(2)} < 1$ at every position? Then the fringes are built photon by photon.

Optional (not needed for the core): Parts E, F and two controls

Polarization is the direction in which the light’s field oscillates; a **polarizer** passes one direction; a **half-wave plate** mirrors the direction about its axis. The pair photons are vertically polarized. **E, quantum eraser** (manual §9.8): a half-wave plate at 22.5° between the two beam splitters makes the polarization diagonal. Arm polarizers crossed at 0° and 90° mark the path: no fringes, $P_d = \frac{1}{4}$ flat. A polarizer at $\pm 45^\circ$ before the detector erases the mark and the fringes return, $P_d = (1 \pm \cos \varphi)/8$ with $\varphi = 4\pi x/\lambda$; with both arm polarizers at 0° the maximum is twice the flat level. **F, Malus’s law for single photons** (manual §9.6): a polarizer at θ from the horizontal passes a heralded photon with probability $\sin^2 \theta$. **Controls.** Manual §9.3, one arm alone (T blocked): *thermal light* (many independent emitters, whose photons bunch) would give 2, but its coherence time (≈ 0.2 ps) is far below Δt , so expect ≈ 1 , not 2. Manual §9.5, GRA with classical light: expect ≈ 1 ; three detectors alone do not make $g^{(2)} < 1$.

What this kit cannot show: entanglement

The course plan promises entangled photons. This kit’s single crystal makes pairs with both photons vertically polarized, $|VV\rangle = |V\rangle \otimes |V\rangle$: “correlated in time and energy (though not polarization-entangled)” (manual Ch. 2). *Entangled* means a two-photon state that cannot be written as $|\alpha\rangle \otimes |\beta\rangle$, such as $|\Phi^+\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$ (H : horizontal). The optional notebook section compares the two with a polarizer on each photon: at the standard angles the CHSH sum S (a combination of four polarizer correlations, defined in the optional notebook section) is $S = \sqrt{2} \approx 1.41$ for $|VV\rangle$ against $2\sqrt{2} \approx 2.83$ for $|\Phi^+\rangle$, while every *local hidden variable model* (each photon leaves the source with ready-made answers for every polarizer angle) obeys $|S| \leq 2$. The Thorlabs *EDU-QOPA1(/M) Polarization-Entanglement Extension Kit* adds a crossed type-I BBO pair plus compensation crystals; it is not used here. *Anchor*: the Micius satellite sent entangled pairs to two ground stations (2017).

Real-World Anchors

A: Hanbury Brown and Twiss (1956) correlated two detectors’ signals, the method behind $g_{\text{HBT}}^{(2)}$; it measured the sizes of stars. **C:** Grangier, Roger and Aspect (1986) ran Part C with heralded photons from atoms and found 0.18; today Quandela builds quantum-dot single-photon sources for photonic quantum computers. **D:** LIGO’s kilometre-long Michelson interferometers reported gravitational waves (2016). **Optional:** Nobel Prize in Physics 2022 (Aspect, Clauser, Zeilinger), entangled photons.

References. Thorlabs, *EDU-QOPA1(/M) Quantum Optics Kit User Guide*, Rev. D (2025-10-25): Ch. 1, 2, 14; §4.2.2, §7.6.4, §9.1–9.8, §11.7, §11.9, §12.2.1. Course document QC10, *Light*. R. Hanbury Brown, R. Q. Twiss, *Nature* **177**, 27 (1956). P. Grangier, G. Roger, A. Aspect, *Europhys. Lett.* **1**, 173 (1986). J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt, *Phys. Rev. Lett.* **23**, 880 (1969). Nobel Prize in Physics 2022 (Aspect, Clauser, Zeilinger). J. Yin et al., *Science* **356**, 1140 (2017). B. P. Abbott et al., *Phys. Rev. Lett.* **116**, 061102 (2016). N. Somaschi et al., *Nat. Photon.* **10**, 340 (2016).

Worksheet — Lab 10: Quantum Optics on the Bench

Group _____ Names _____ Date _____

W1 · Part A: attenuated laser (HBT), kit setting, 20 s per run. $\Delta t =$ _____ ns. Rates in Hz; accidentals = $R_A R_B \Delta t$; $g_{\text{HBT}}^{(2)} = R_{AB} / \text{accidentals}$.

run	R_A	R_B	R_{AB}	accidentals	$g_{\text{HBT}}^{(2)}$	predicted
run 1						1
run 2						1
run 3						1
<i>Optional: extra ND filter (only if i-LAMP has one; tens of minutes per run)</i>						
extra ND filter						1

Compare: Table 1 gives 1.0020. Your runs: consistent with 1 and with each other? _____

W2 · Part B: pair source. $g_{\text{PS}}^{(2)} = R_{TA} / (R_T R_A \Delta t)$.

	R_T	R_A	R_{TA}	$R_T R_A \Delta t$	$g_{\text{PS}}^{(2)}$	predicted
your run						≥ 30

Compare: at least 30? Table 2 gives ≈ 115 . _____

W3 · Part C: heralded photons (GRA). $g_{\text{GRA}}^{(2)} = N_{TAB} N_T / (N_{TA} N_{TB})$; $\sigma \approx g_{\text{GRA}}^{(2)} / \sqrt{N_{TAB}}$. The Table 2 row is the manual's example.

	time (s)	N_T	N_{TA}	N_{TB}	N_{TAB}	$g_{\text{GRA}}^{(2)}$	σ	$(1 - g^{(2)}) / \sigma$
Table 2	20	4.638 M	385865	324688	513	0.0190	0.00084	1170
your run								
predicted	one photon cannot take both exits					< 1		

Accidentals (the window box). Rates $R_X = N_X / \text{time}$ in Hz; $R^{(3)} = R_T R_A R_B \Delta t^2$; $R^{(2+1)} = (R_{TA} R_B + R_{TB} R_A) \Delta t$; $g_{\text{acc}}^{(2)} = (R^{(3)} + R^{(2+1)}) R_T / (R_{TA} R_{TB})$; chance triples = $(R^{(3)} + R^{(2+1)}) \times \text{time}$.

	R_T	R_A	R_B	R_{TA}	R_{TB}
Table 2	231900	145250	146250	19293.25	16234.4
your run					

	$R^{(3)}$ (Hz)	$R^{(2+1)}$ (Hz)	$g_{\text{acc}}^{(2)}$	chance triples	$\frac{\text{chance} - N_{TAB}}{\sqrt{N_{TAB}}}$
Table 2	0.1232	25.898	0.0193	520	0.33
your run					

Compare: is $g_{\text{GRA}}^{(2)}$ below 1 by many σ ? Do chance triples explain your N_{TAB} ? _____

W4 · Part D: heralded Michelson. Positions x of the fringe maxima of R_{TB} (in μm), in order; one maximum per fringe. Record at least the first and the last, and count them all for N_{max} .

maximum	1	2	3	4	5	6	7	8	9	10
x (μm)										
maximum	11	12	13	14	15	16	17	18	19	20
x (μm)										

	N_{max}	x_{first}	x_{last}	spacing (nm)	λ (nm)	R_{max}	R_{min}	$\mathcal{V}$
your scan								
predicted				≈ 405	≈ 810			1 (ideal)

$\lambda = 2(x_{\text{last}} - x_{\text{first}})/(N_{\text{max}} - 1)$. $g_{\text{GRA}}^{(2)}$ at a maximum _____, a minimum _____, in between _____ (predicted: all < 1).

Compare: is λ within 3% of 810 nm? _____

W5 · Optional parts.

F: polarizer θ	0°	30°	45°	60°	90°	
predicted $\sin^2 \theta$	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1	
measured (relative)						

E: eraser	arms crossed	crossed + $\pm 45^\circ$	both arms 0°
predicted	no fringes	fringes	fringes
observed			

controls	one arm alone (§9.3)	classical GRA (§9.5)
predicted $g^{(2)}$	≈ 1	≈ 1
measured $g^{(2)}$		